

In Utero and Childhood Polybrominated Diphenyl Ether (PBDE) Exposures and Neurodevelopment in the CHAMACOS Study

background: California children's exposures to polybrominated diphenyl ether flame retardants (PBDEs) are among the highest worldwide. PBDEs are known endocrine disruptors and neurotoxicants in animals. Objective: Here we investigate the relation of in utero and child PBDE exposure to neurobehavioral development among participants in CHAMACOS (Center for the Health Assessment of Mothers and Children of Salinas), a California birth cohort. Methods: We measured PBDEs in maternal prenatal and child serum samples and examined the association of PBDE concentrations with children's attention, motor functioning, and cognition at 5 (n = 310) and 7 years of age (n = 323). Results: Maternal prenatal PBDE concentrations were associated with impaired attention as measured by a continuous performance task at 5 years and maternal report at 5 and 7 years of age, with poorer fine motor coordination—particularly in the nondominant—at both age points, and with decrements in Verbal and Full-Scale IQ at 7 years. PBDE concentrations in children 7 years of age were significantly or marginally associated with concurrent teacher reports of attention problems and decrements in Processing Speed, Perceptual Reasoning, Verbal Comprehension, and Full-Scale IQ. These associations were not altered by adjustment for birth weight, gestational age, or maternal thyroid hormone levels. Conclusions: Both prenatal and childhood PBDE exposures were associated with poorer attention, fine motor coordination, and cognition in the CHAMACOS cohort of school-age children. This study, the largest to date, contributes to growing evidence suggesting that PBDEs have adverse impacts on child neurobehavioral development.